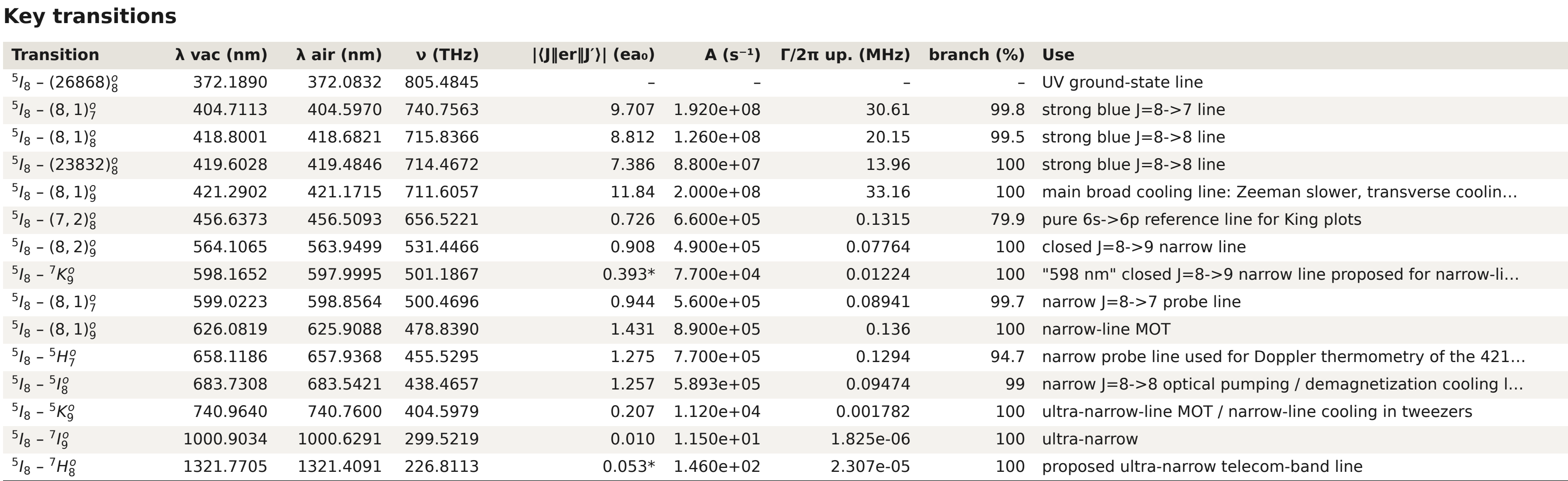
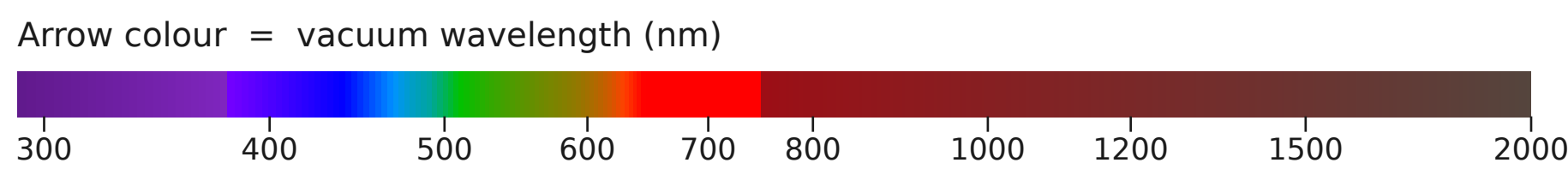


¹⁶³Dy (Dy I), nuclear spin $I = 5/2$. Energy levels and transitions below 2 μm : 39 levels, 55 lines.
Dy⁺ ionisation limit 47901.76 cm^{-1} = 5.93906 eV (208.761 nm)

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Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| er \| J' \rangle |$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Dy hyperfine structure ($\eta = 5/2$)									
Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)						
$5f_6$	162.754(20)	1152.87(4)	F=11/2: -3251.26	F=13/2: -2549.30	F=15/2: -1588.04	F=17/2: -302.63	F=19/2: +1380.44	F=21/2: +3543.30	
$7f_{5/2}^o$	131.000	1901.000	F=11/2: -2270.27	F=13/2: -2005.70	F=15/2: -1450.93	F=17/2: -499.01	F=19/2: +971.23	F=21/2: +3095.25	
$7f_6$	125.000	2901.000	F=13/2: -2129.56	F=15/2: -2087.96	F=17/2: -1654.01	F=19/2: -682.66	F=21/2: +988.20	F=23/2: +3537.75	
$5k_{5/2}^o$	142.91(3)	4105.02(0)	F=13/2: -2164.17	F=15/2: -2360.07	F=17/2: -2034.75	F=19/2: -982.97	F=21/2: +1024.68	F=23/2: +4241.73	
$5f_{5/2}^o$	152.560(20)	2357.01(0)	F=11/2: -2592.92	F=13/2: -2329.00	F=15/2: -1715.13	F=17/2: -618.71	F=19/2: +1110.50	F=21/2: +3640.45	
(8.1) $_{1/2}^o$	261.700	-334.000	F=13/2: -6657.11	F=15/2: -4591.21	F=17/2: -2294.39	F=19/2: -216.64	F=21/2: +2923.23	F=23/2: +5804.75	
(8.1) $_{3/2}^o$	89.858(17)	943.21(9)	F=9/2: -1444.76	F=11/2: -1241.27	F=13/2: -879.52	F=15/2: -298.87	F=17/2: +570.65	F=19/2: +2103.32	
(8.2) $_{1/2}^o$	207.295(13)	1652.21(9)	F=13/2: -4615.44	F=15/2: -3570.97	F=17/2: -2166.94	F=19/2: -320.74	F=21/2: +2059.95	F=23/2: +5077.19	
(8.1) $_{3/2}^o$	121.620(20)	1844.94(0)	F=13/2: -2407.45	F=15/2: -2065.05	F=17/2: -1431.00	F=19/2: -413.08	F=21/2: +1091.83	F=23/2: +3197.68	
(26868) $_{1/2}^o$	110.31(2)	1595(57)	F=11/2: -191.533	F=13/2: -1689.04	F=15/2: -1220.66	F=17/2: -418.69	F=19/2: -818.57	F=21/2: +2604.75	

Measured A, B constants; sources in the data file

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^1S_0 - ^3P_0$ (2686.8) μ m	372.189	164-163	-44(10)
$^1S_0 - ^3P_0$ (2686.8) μ m	372.189	164-162	-178(13)
$^1S_0 - ^3P_0$ (2686.8) μ m	372.189	164-161	-194(13)
$^1S_0 - ^3P_0$ (2686.8) μ m	372.189	164-160	-339(13)
$^1S_0 - (^8, ^7)_1$	421.290	164-163	-616(35)
$^1S_0 - (^8, ^7)_1$	421.290	164-162	-913.2(8)
$^1S_0 - (^8, ^7)_1$	421.290	164-161	-1635.0(10)
$^1S_0 - (^8, ^7)_1$	421.290	164-160	-1895.0(20)
$^1S_0 - (^8, ^7)_1$	421.290	164-158	-2868(9)
$^1S_0 - (^8, ^7)_1$	421.290	164-156	-4300(15)
$^1S_0 - (^7, ^2)_1$	456.637	164-163	-660(3)
$^1S_0 - (^7, ^2)_1$	456.637	164-162	-971.0(20)
$^1S_0 - (^7, ^2)_1$	456.637	164-161	-1744(3)
$^1S_0 - (^7, ^2)_1$	456.637	164-160	-2020(3)
$^1S_0 - (^7, ^2)_1$	456.637	164-158	-3061(4)
$^1S_0 - (^7, ^2)_1$	456.637	164-156	-4604(5)
$^1S_0 - (^8, ^2)_1$	564.106	164-163	-685.0(20)
$^1S_0 - (^8, ^2)_1$	564.106	163-162	-328(3)
$^1S_0 - (^8, ^2)_1$	564.106	162-161	-801(3)
$^1S_0 - (^8, ^2)_1$	564.106	161-160	-284.0(20)
$^1S_0 - (^8, ^2)_1$	564.106	164-162	-1012.0(20)
$^1S_0 - (^8, ^2)_1$	564.106	162-160	-1086(5)

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$